Decoding Parameter and Sampling Analysis

Group Name: Attention is All We Need

Group Members

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1. Introduction

Decoding parameters influence how a language model selects its next token. Changing them can alter response wording, length, diversity, factual content, and compliance with structured-output instructions. These effects matter when a local model is used for research, analysis, or automation.

This project addresses a practical question: how sensitive are two small local language models to controlled changes in temperature, top-k, and top-p when the prompt and model are held constant? The study focuses on output sensitivity rather than repeated-run determinism.

2. Background and Technical Overview

Temperature changes the sharpness of the probability distribution used for token selection. Lower values favor higher-probability tokens, while higher values generally allow more variation. Top-k restricts selection to the k most probable candidate tokens. Top-p, also called nucleus sampling, restricts selection to the smallest candidate set whose cumulative probability reaches p. Their effects depend on the model and on the other decoding settings.

The benchmark uses the exact local Ollama models Qwen2.5:3b and Llama3.2:3b. The report makes no additional architecture or capability claims beyond the verified model tags and recorded local execution.

3. Methodology

In plain terms, the experiment starts with a baseline setting and changes one decoding parameter at a time. Every prompt is sent once to each model under each setting. This isolates baseline-relative observations but does not estimate interactions among parameters or run-to-run variation.

keywords:

3.1 Prompts and configurations

ID	Objective	Exact prompt
D1	variability	Explain how historical electricity use, weather, and calendar effects help forecast demand. State two benefits and one limitation.
F1	factual	Explain MAE, RMSE, and MAPE in forecasting. State how each is calculated, how RMSE differs from MAE, and one limitation of MAPE.
S1	structure	Return only valid JSON with keys actual, predicted, and error. Actual demand is 500 MW and predicted demand is 450 MW. Set error to the absolute difference. Use numbers, not strings.

Configuration	Temperature	Top-k	Top-p
baseline	0.0	1	0.3
temperature_medium	0.5	1	0.3
temperature_high	1.2	1	0.3
top_k_medium	0.0	15	0.3
top_k_high	0.0	50	0.3
top_p_high	0.0	1	1.0

There were 3 prompts × 6 configurations × 2 models = 36 generations.

Temperature comparisons changed only temperature. Top-K comparisons changed only Top-K. The Top-P comparison changed only Top-P. However, Top-K was fixed at 1 in the temperature and Top-P comparisons. Top-K = 1 may restrict the candidate tokens so strongly that temperature or Top-P has little practical effect. These are therefore configuration-level observations, not perfect isolated tests of each parameter.

The token proxy counted word-like tokens plus punctuation with this regular expression: `\\b[\w'-]+\b|[^\w\s]`. It is not a count from the model's own tokenizer.

4. Results

The tables below use the saved evaluated records. The full response text is available in the separate document Actual_Model_Outputs_Readable.docx.

4.1 D1: output length and topic coverage

Qwen's baseline output had 351 words and 465 token-proxy units. Every Qwen alternative had 348 words. Llama's baseline had 334 words and every Llama alternative had 320 words. All 12 D1 outputs covered 5/5 required topics.

Model	Configuration	Words	Token proxy	Topics
qwen2.5:3b	Baseline	351	465	5/5
qwen2.5:3b	Temperature medium	348	467	5/5
qwen2.5:3b	Temperature high	348	467	5/5
qwen2.5:3b	Top-K medium	348	467	5/5
qwen2.5:3b	Top-K high	348	467	5/5
qwen2.5:3b	Top-P high	348	467	5/5
llama3.2:3b	Baseline	334	432	5/5
llama3.2:3b	Temperature medium	320	413	5/5
llama3.2:3b	Temperature high	320	413	5/5
llama3.2:3b	Top-K medium	320	413	5/5
llama3.2:3b	Top-K high	320	413	5/5
llama3.2:3b	Top-P high	320	413	5/5

Table 3. D1 length and topic coverage.

TF-IDF similarity ranged from 0.66 to 1.00 for Qwen and from 0.92 to 1.00 for Llama. Higher values mean more similar wording. This is a wording measure, not a direct test of determinism.

4.2 F1: preliminary factual-content checklist

The F1 evaluator looked for five expected concepts: MAE definition, RMSE definition, the RMSE–MAE difference, MAPE definition, and the MAPE zero or near-zero limitation. These scores show lexical detection only. They do not prove factual accuracy, completeness, or absence of hallucinations.

Model	Configuration	Checklist score	Status
qwen2.5:3b	Baseline	1.00	Needs human review

qwen2.5:3b	Temperature medium	0.80	Needs human review
qwen2.5:3b	Temperature high	0.80	Needs human review
qwen2.5:3b	Top-K medium	0.80	Needs human review
qwen2.5:3b	Top-K high	0.80	Needs human review
qwen2.5:3b	Top-P high	0.80	Needs human review
llama3.2:3b	Baseline	0.80	Needs human review
llama3.2:3b	Temperature medium	1.00	Needs human review
llama3.2:3b	Temperature high	1.00	Needs human review
llama3.2:3b	Top-K medium	1.00	Needs human review
llama3.2:3b	Top-K high	1.00	Needs human review
llama3.2:3b	Top-P high	1.00	Needs human review

Table 4. F1 preliminary checklist results.

Qwen scored 1.00 at baseline and 0.80 for every alternative because the RMSE definition was not detected by the lexical checklist in the alternatives. Llama scored 0.80 at baseline and 1.00 for every alternative because the MAPE zero/near-zero limitation was not detected at baseline. These are non-detection labels, not claims that the model's statements were false.

4.3 S1: JSON structure

Model	Configurations passed	Configurations failed	Interpretation
Qwen 2.5:3B	6/6	0/6	All tested outputs passed strict JSON checks
Llama 3.2:3B	0/6	6/6	No tested output passed strict JSON checks

Table 5. S1 strict JSON results.

The strict check required valid JSON, exactly the keys actual, predicted, and error, numeric values, the expected numbers, and no extra text. This result applies only to this prompt and these saved generations.

5. What changed across configurations?

The following table gives the actual baseline-relative D1 changes. Because all five alternatives within a model had the same summary values, the same change appears for each alternative.

Model	Changed configuration	Length change	Coverage change
Qwen 2.5:3B	Temperature medium	-3 words	unchanged at 5/5
Qwen 2.5:3B	Temperature high	-3 words	unchanged at 5/5
Qwen 2.5:3B	Top-K medium	-3 words	unchanged at 5/5
Qwen 2.5:3B	Top-K high	-3 words	unchanged at 5/5
Qwen 2.5:3B	Top-P high	-3 words	unchanged at 5/5
Llama 3.2:3B	Temperature medium	-14 words	unchanged at 5/5
Llama 3.2:3B	Temperature high	-14 words	unchanged at 5/5

Llama 3.2:3B	Top-K medium	-14 words	unchanged at 5/5
Llama 3.2:3B	Top-K high	-14 words	unchanged at 5/5
Llama 3.2:3B	Top-P high	-14 words	unchanged at 5/5

Table 6. D1 changes from baseline.

The non-baseline D1 records differed from the baseline in measured length. However, because all five alternative configurations produced identical summary metrics within each model, the observed difference cannot be attributed to any individual decoding parameter.

For F1, Qwen alternatives were -0.20 from baseline and Llama alternatives were +0.20. For S1, Qwen stayed at strict pass and Llama stayed at strict fail. These patterns are observations from one generation per condition.

6. Direct model comparison

Question	Observed answer
Did both models cover D1 topics?	Yes. Both covered 5/5 in every condition.
Did both models have the same D1 length?	No. Qwen was 351 baseline / 348 alternatives; Llama was 334 / 320.
Did both models pass S1?	No. Qwen passed 6/6; Llama passed 0/6.
Can F1 scores prove factual accuracy?	No. They are preliminary lexical checklist results.
Can one parameter be blamed for D1 length changes?	No. Alternative configurations shared identical summary metrics.

Table 7. Plain-language comparison of the main findings.

8. Conclusion

This experiment shows how the two tested local models behaved on three specific prompts and six configurations. All D1 outputs covered the requested topics. Qwen produced valid expected JSON in every S1 condition, while Llama did not. D1 length differed between baseline and non-baseline records, but identical summary results across alternatives prevent attribution to an individual decoding parameter. F1 results remain preliminary and require human review.

9. Figures

These figures are descriptive summaries. They do not prove causation, determinism, statistical significance, or generalization.

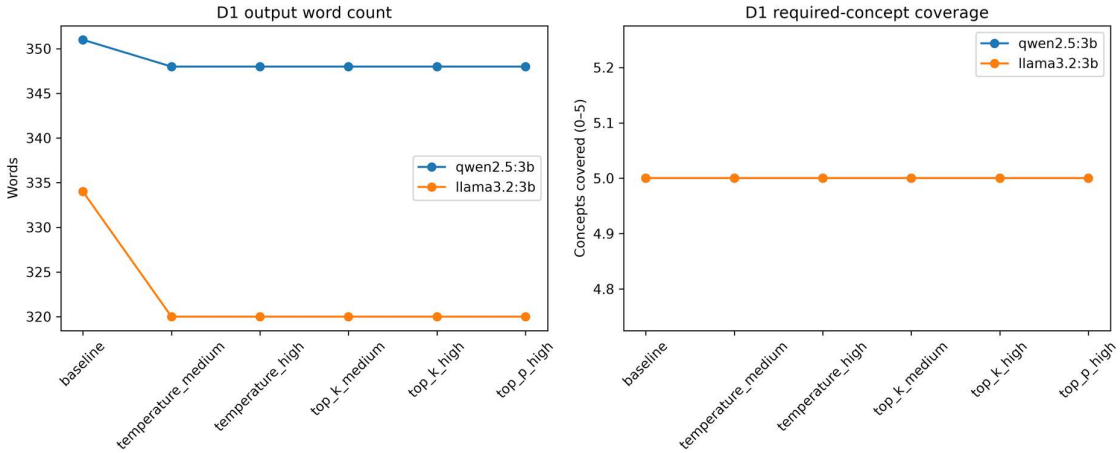


Figure 1. D1 output length and topic coverage.

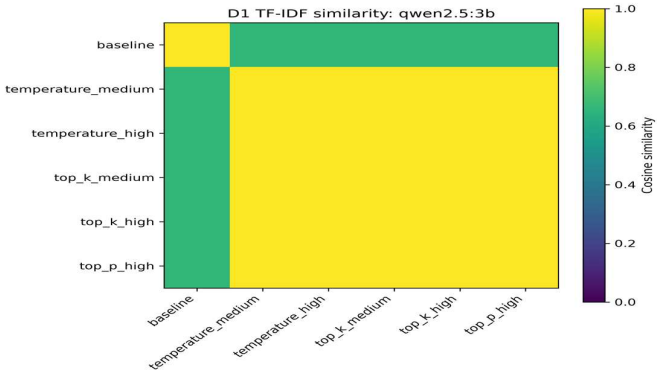


Figure 2. D1 wording similarity for Qwen.

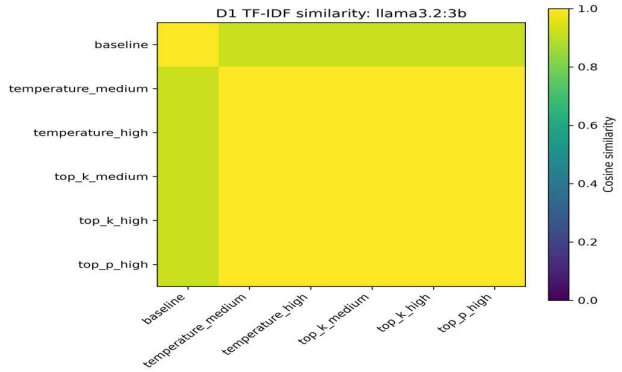


Figure 3. D1 wording similarity for Llama.

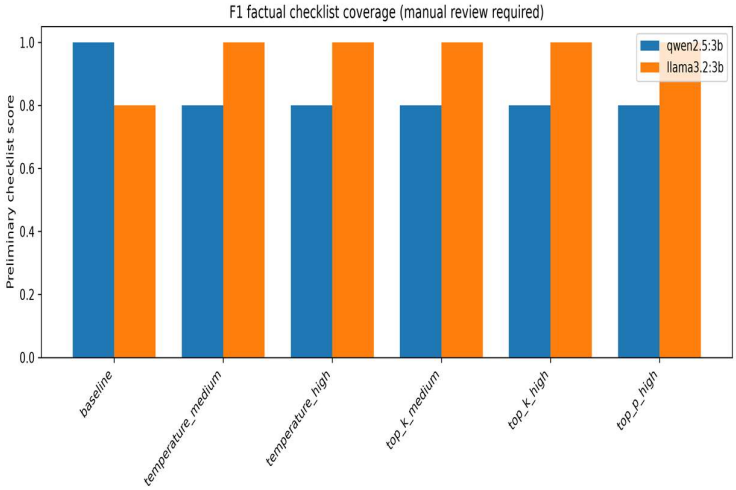


Figure 4. F1 preliminary checklist scores.

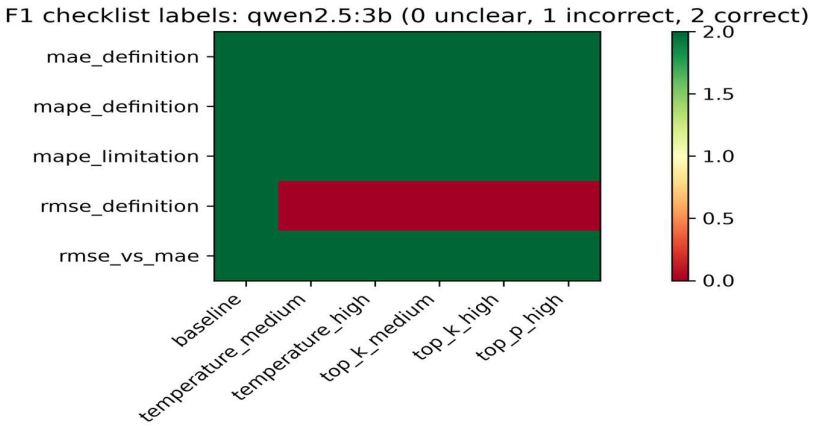


Figure 5. F1 checklist results for Qwen.

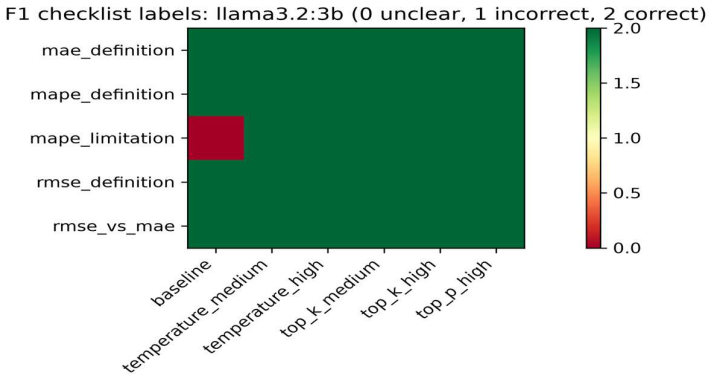


Figure 6. F1 checklist results for Llama.

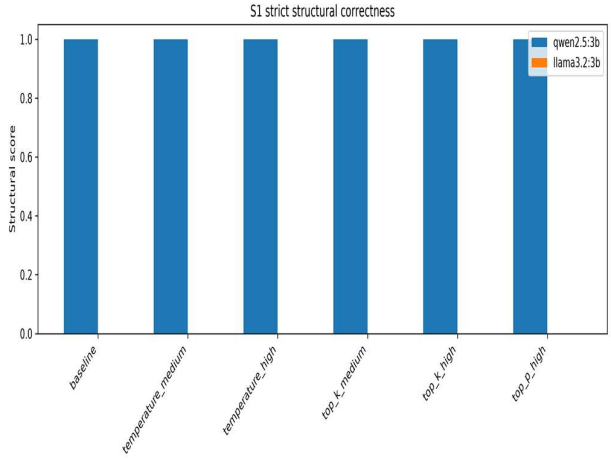


Figure 7. S1 structural scores.

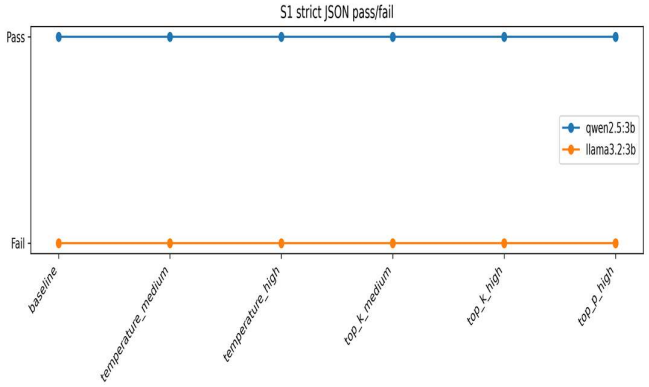


Figure 8. S1 strict JSON pass/fail.

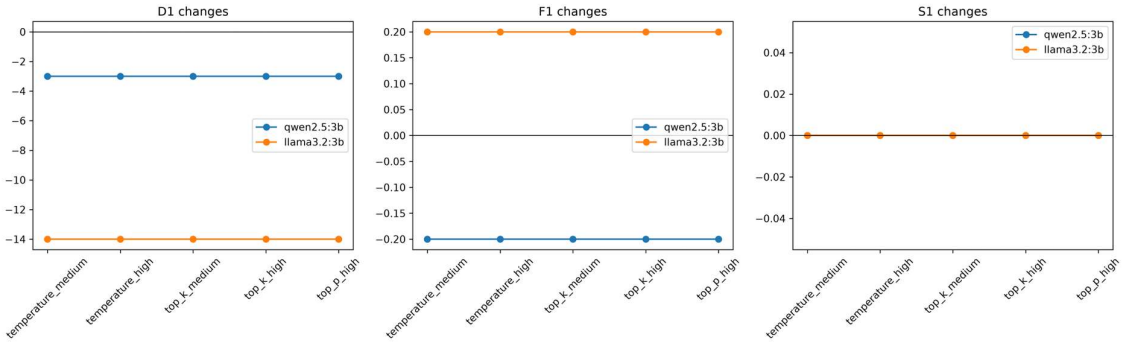


Figure 9. Baseline-relative metric changes.